# State Estimation and Filtering for Robotic Perception

**AA273**

Emerging topics: learning based robot autonomy,

3DGS, 3D world models

Stanford University

MSL

# AA273: State Estimation and Filtering for Robotic Perception

**Announcements:**

- HW 6 due Fri Mar 6
- Project Reports: due Tue Mar 17
  - 5-6 pages + 1 page per group member > 1
  - Latex/Overleaf, IEEE conf format
  - >= 10 refs
  - Describe research and results
  - Re-use proposal material
- Project Presentations: 3/10, 3/12
  - 4 min pres + 1 min Q and A
  - Sign up for preferred time slot
  - Share slides through zoom meeting
  - Present on your computer

**Lecture 18:**

- learning based robot autonomy
- 3DGS
- 3D World models

# Humanoids

Unitree H2

**Terrifying? Maybe. Impressive? Definitely! But…**

# AVs

Waymo

**Impressive!** **But actually uses classical robot autonomy stack.**

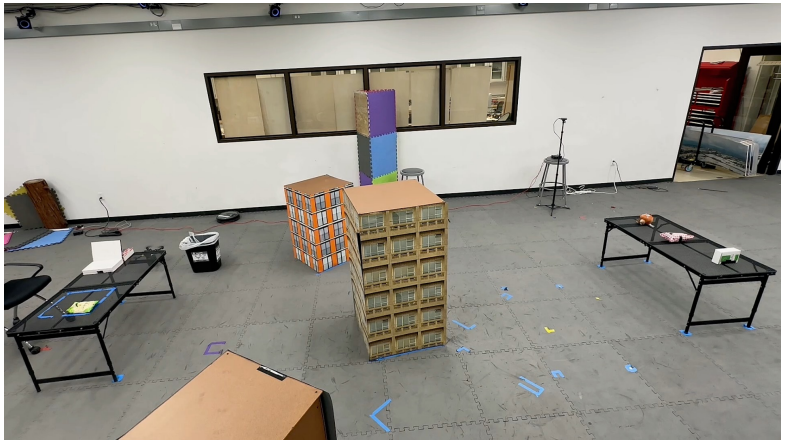
# AI-Based Manipulation

Physical Intelligence (PI)

**Not as flashy? Maybe. But this is a paradigm shift. Fundamentally new methods and capabilities.**

# Drones

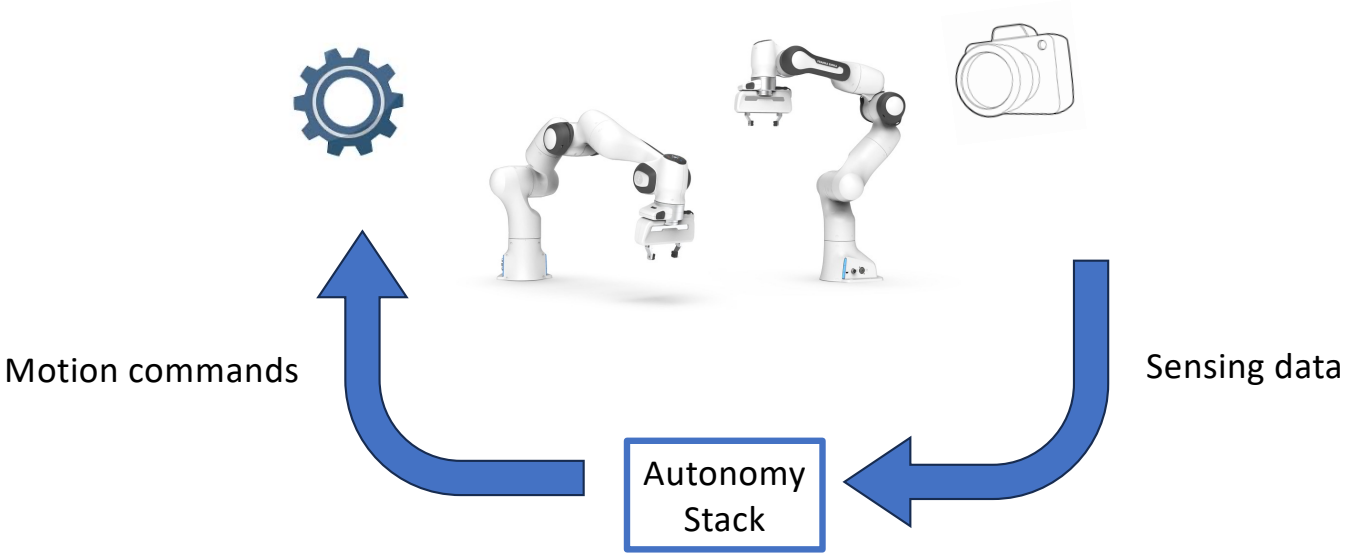
Prof. Davide Scarramuza, U. Zurich
Mostly uses classical autonomy tools.



Pi in the Sky, Stanford MSL (my lab)
End-to-end AI based tools. Can respond to language commands and operate in open-ended real world spaces.

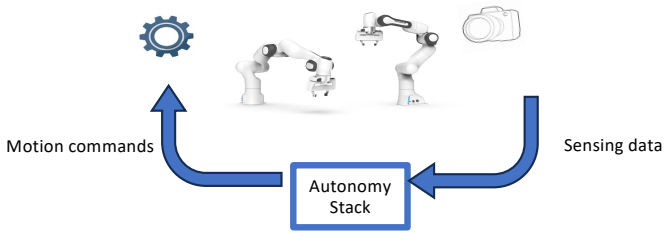
# Robot Autonomy

Algorithms and software stacks to empower robots to make their own decisions, with little or no human intervention.



# AI-Driven Robot Autonomy

Algorithms and software stacks to empower robots to make their own decisions, with little or no human intervention.




## Classical Autonomy

- Since 1980s
- Modular, algorithmic tools
- Limited machine learning
- Focus on physics-driven modeling, and mathematical performance guarantees

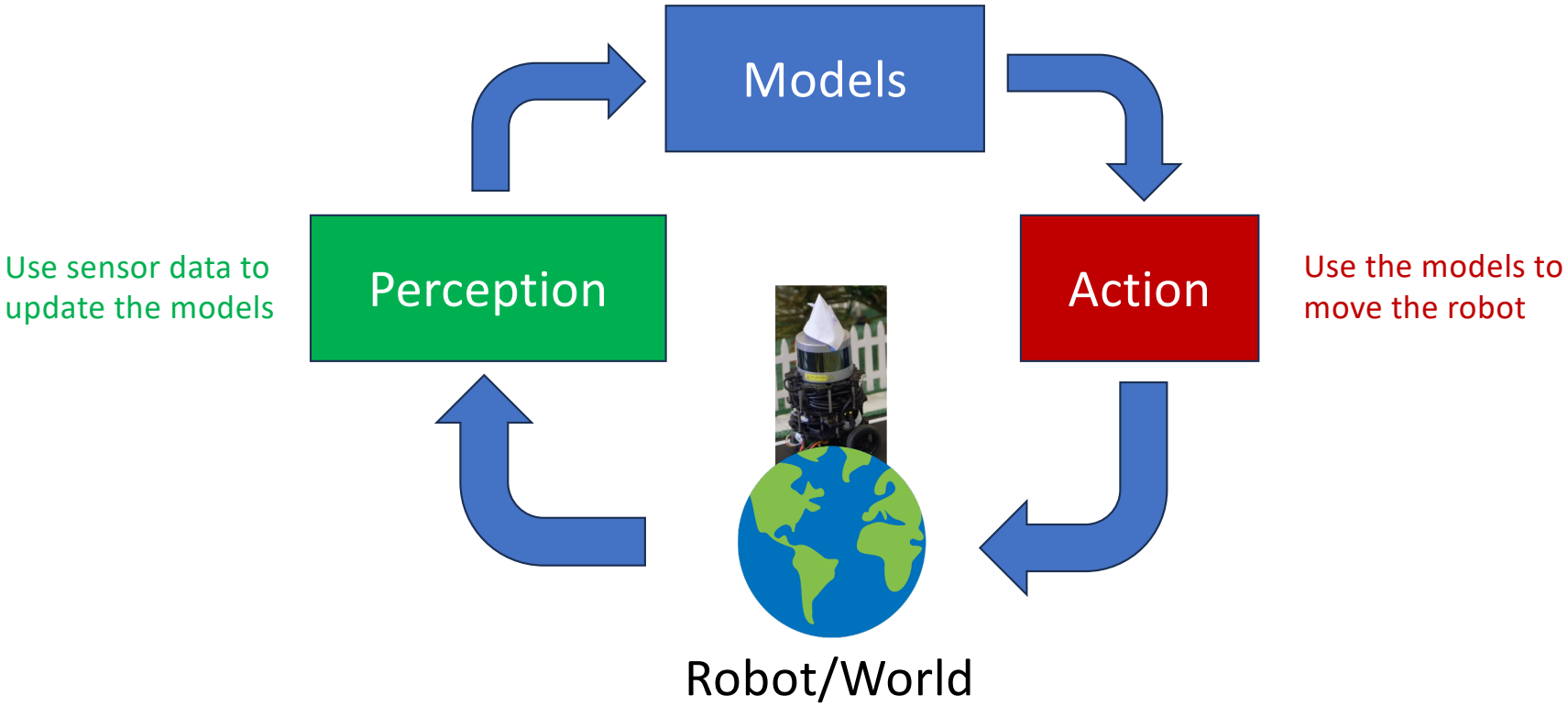
## AI-Driven Autonomy

- Since approx. 2022
- Heavy use of machine learning, end-to-end policies
- Focus on data, often from human demonstration
- Incorporates AI architectures or pre-trained models
- Language commanded, vision-based, generalist

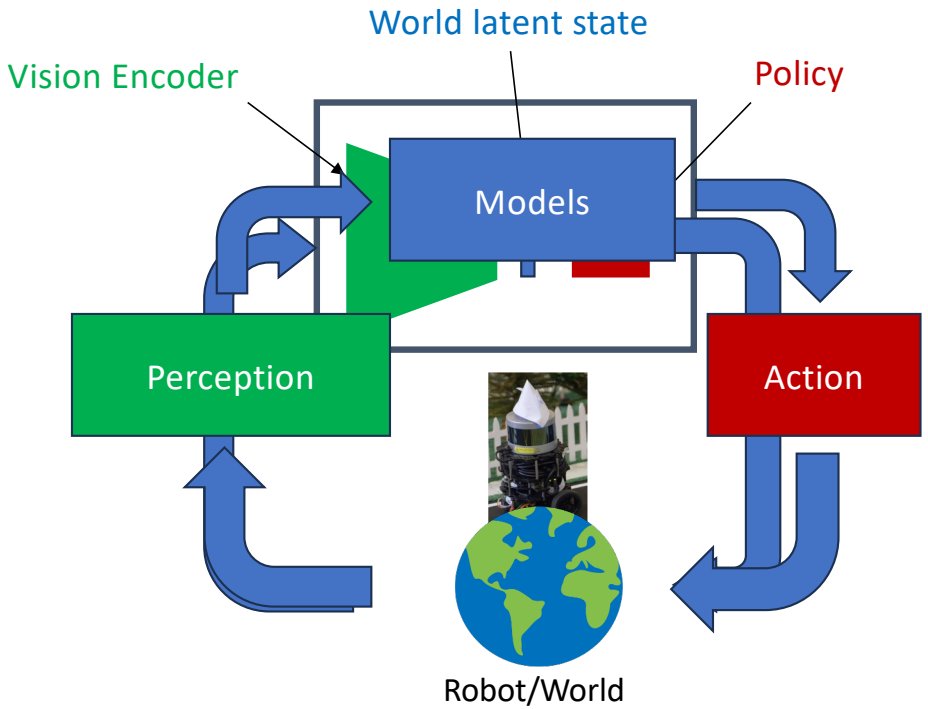
# Roadmap

- Examples of AI-Driven Robot Autonomy
- Robot autonomy policies
  - The robot autonomy stack
  - Imitation Learning (IL)
  - Reinforcement Learning (RL)
- World Models, 3D Models, and Simulators
- Challenges and opportunities

# Classical Autonomy Stack: the Perception-Action Loop



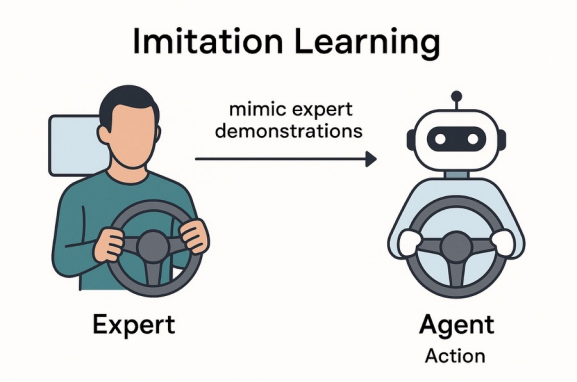
# AI-Based Autonomy Stack




End-to-End Visuo-Motor Policy

- Still have the three basic components internally in learning architecture
- Imitation learning (IL)/Behavior Cloning (BC)
  - Learn visuomotor policy from human demonstrations
  - Very successful in recent research!
- Reinforcement learning (RL)
  - Usually learned in simulators
  - Quite data-hungry
  - Sim-to-real is often a major challenge

# Imitation Learning (IL) vs Reinforcement Learning (RL)

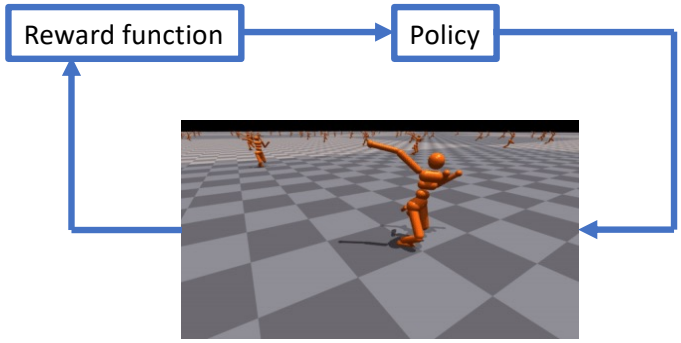



Uses real-world data from a human operating the robot.

**Pros:** It works! Key approach behind recent AI autonomy breakthroughs.

**Cons:** It requires human-intensive, expensive, data collection.

Reinforcement Learning



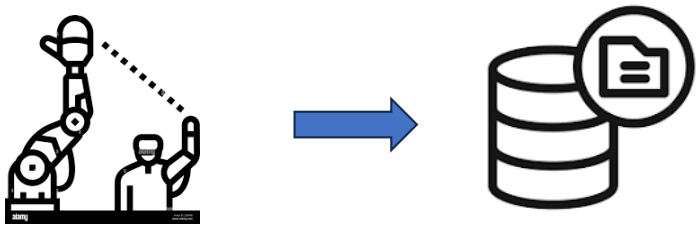

Simulator

Uses data generated from trial and error in a simulator.

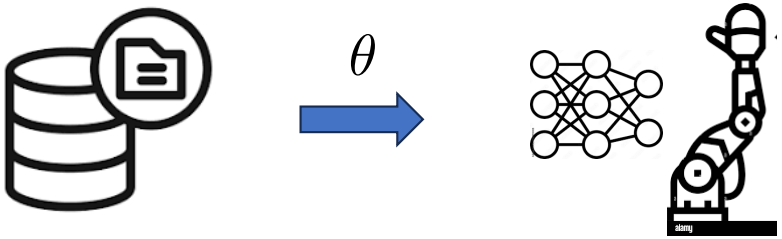
**Pros:** No real world data, just a simulator.

**Cons:** It (usually) only works for small proprioceptive policies (deaf, dumb, and blind). Even still sim-to-real gap is a major challenge

# Imitation Learning Framework

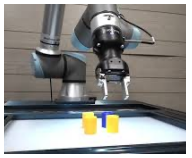


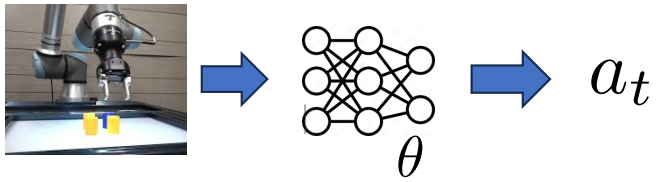
Human teleoperation to collect demonstration data




Data used to train weights of a neural network policy to drive the robot autonomously

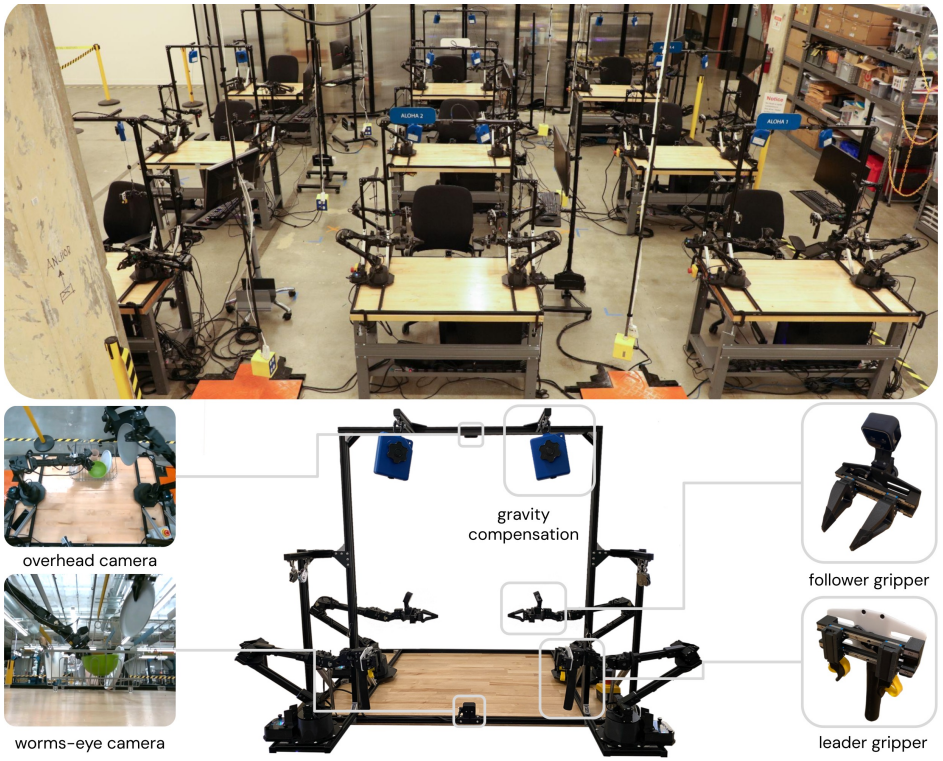
Data is usually image/action pairs

$$\left( \text{  } , a_t \right)$$




Trained Visuo-motor policy

# Human teleoperation data collection: ALOHA



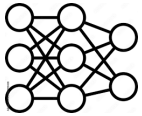
https://aloha-2.github.io/

https://news.stanford.edu/stories/2024/04/meet-robot-that-can-saute-shrimp

# Imitation Learning: Policy Training

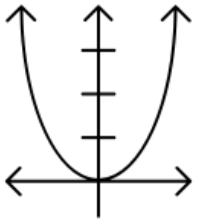
$\{(I_i, a_i)\}_{i=1}^{N}$

Dataset with N
Image-action pairs



$a_t = \pi_\theta(I_t)$

Policy model with
free parameters $\theta$



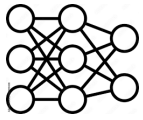
$\text{dist}(a_t, \pi_\theta(I_t))$

Loss function to penalize
bad action prediction

$$\min_\theta \sum_{i=1}^{N} \text{dist}(a_i, \pi_\theta(I_i))$$

Training objective function

PyTorch



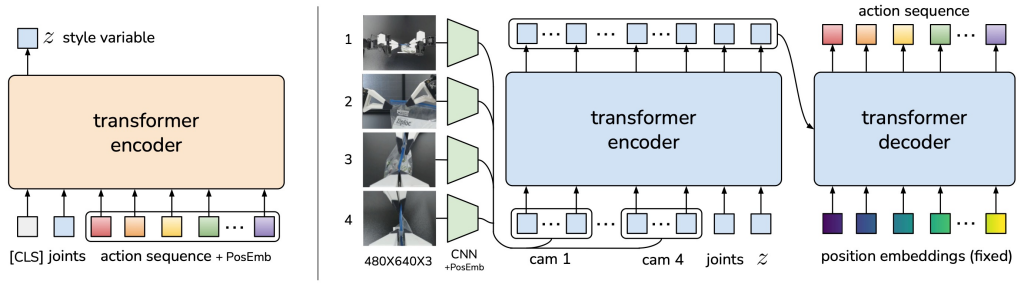
$a^* = \pi_{\theta^*}(I)$

Optimized or "trained"
policy with parameters $\theta^*$

**Advantages:**
(i) No task specification
(ii) No simulation
(iii) All real-world, real data

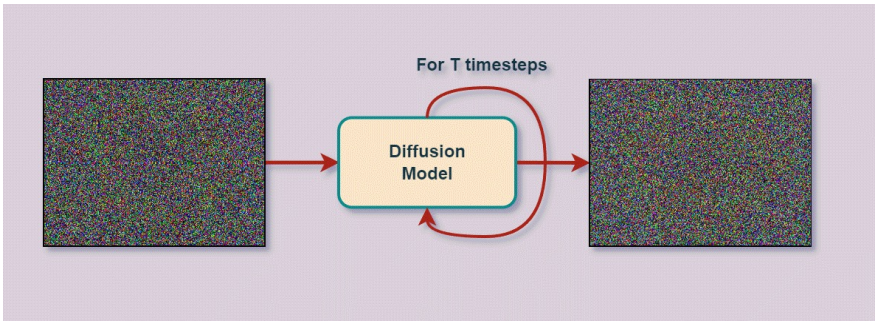
# Policy Architectures

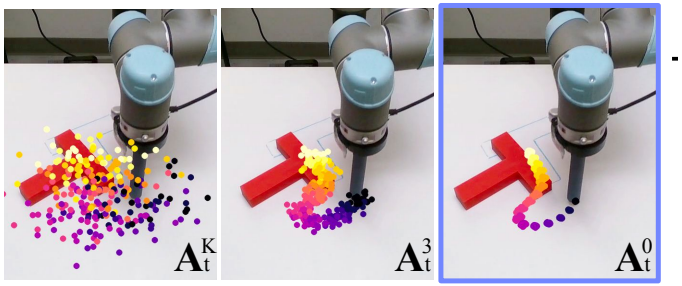
ACT (Action Chunking Transformer): transformer architecture, borrowed from LLMs




From ACT paper: Zhao et al.

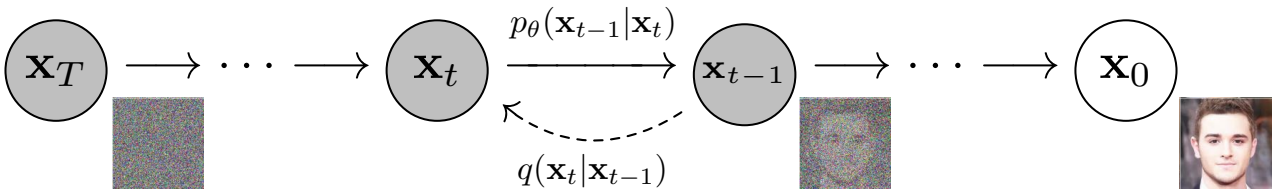
(DP) Diffusion Policy: denoise diffusion model, borrowed from image generation




From DP paper: Chi et al.

# Denoising Diffusion Probabilistic Models




Credit: DDPS paper Ho. Et al.

Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." *Advances in neural information processing systems* 33 (2020): 6840-6851. (36k citations)

Song, Jiaming, Chenlin Meng, and Stefano Ermon. "Denoising diffusion implicit models." *arXiv preprint arXiv:2010.02502* (2020). (13k citations)

Reverse process (learned):

$$p_\theta(\mathbf{x}_{0:T}) := p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t), \qquad p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$

Forward process (known):

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) := \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}), \qquad q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

$$X_t = \sqrt{1-\beta_t} X_{t-1} + \sqrt{\beta_t}\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I)$$

# Denoising Diffusion Probabilistic Models

Credit: DDPS paper Ho. Et al.

Write forward process as SS model:

$$X_t = \sqrt{1-\beta_t} X_{t-1} + \sqrt{\beta_t}\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I)$$

define: $\alpha_t = 1 - \beta_t$

and: $\overline{\alpha}_t = \prod_{\tau=1}^{t} \alpha_\tau$

We get:

$$X_t = \sqrt{\overline{\alpha}_t} X_0 + \sqrt{1-\overline{\alpha}_t}\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I)$$

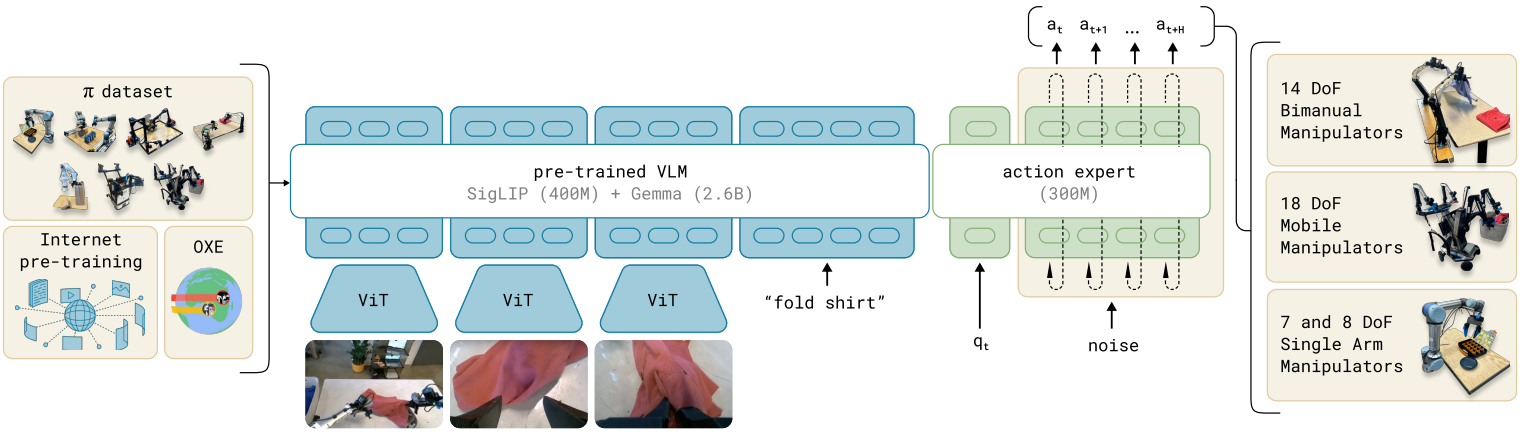
**Algorithm 1** Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$
6: **until** converged

**Algorithm 2** Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

# The Vision Language Action (VLA) Paradigm




From Pi0 paper: Black et al.

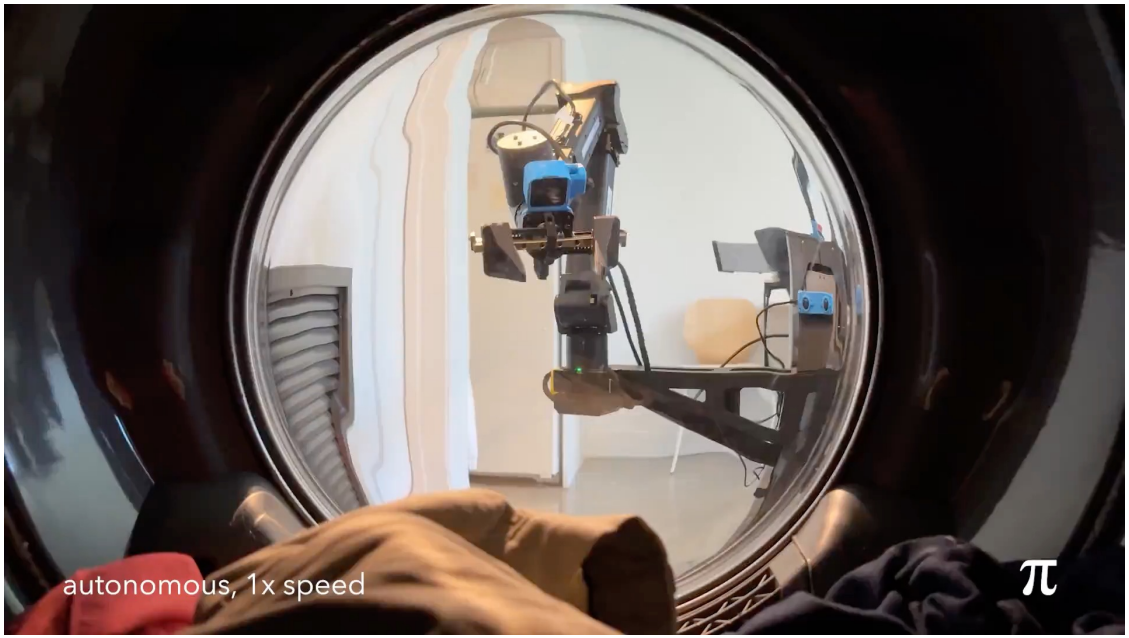
Prime example: Physical Intelligence (PI)

2022 RT-1
2023 Palm-e
2023 RT-2

2024 Octo
2024 OXE/RT-X
2024 OpenVLA

2024 Pi_0
2025 Pi_0.5
2025 GR00T

# Pi0: The leading example VLA




Physical Intelligence (PI)

**Very impressive!**

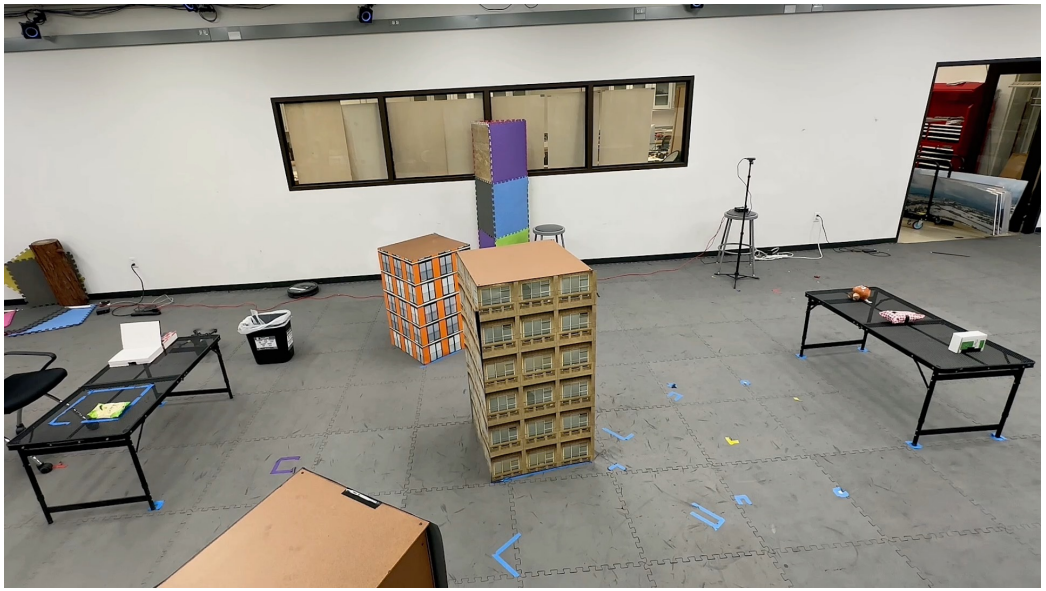
But this is just the start…

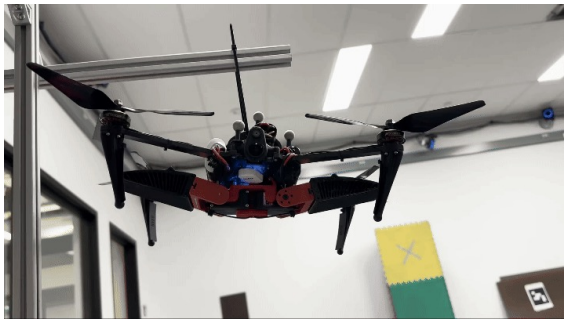
- Trained from a VLM foundation model (Google's PaliGemma)
- Does respond to human language
- Does learn from human demonstrations

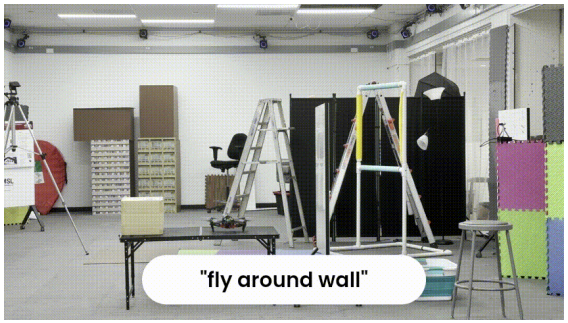
**Limitations**

- Not very generalizable, overfits to training tasks
- Does not "understand" language, poor vision-action-language grounding
- Does not improve through experience
- Is not interactive with other robots or humans

# Pi in the Sky







Pi in the Sky, Stanford MSL (my lab)
End-to-end AI based tools. Can respond to language commands and operate in open-ended real world spaces.

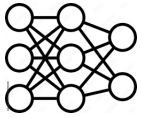
# Human-Piloted Manipulation Demonstrations

# Reinforcement Learning (RL) Framework

(i) Reward usually hand engineered.

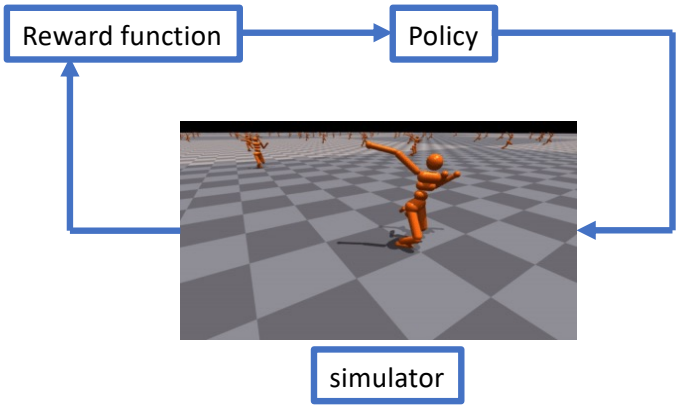
(ii) Recent successes use human demonstrations as reward template.

$r(x_t, a_t)$



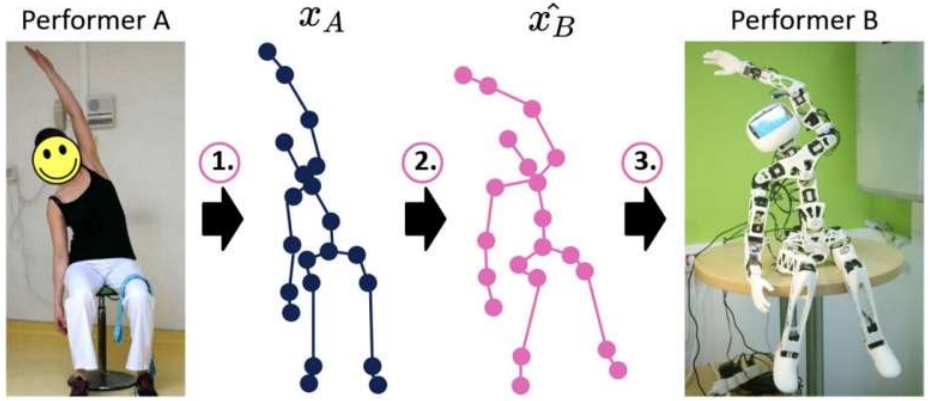
$a_t = \pi_\theta(x_t)$

Policy model with free parameters. State based, not image based.

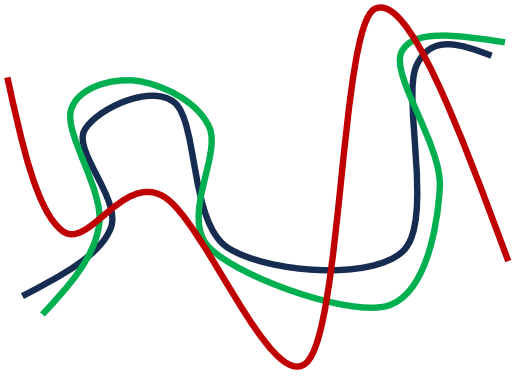



Sim-to-real transfer

# RL + Human Motion Template




$r(x_t, a_t)$



Human template

High reward value

Low reward value

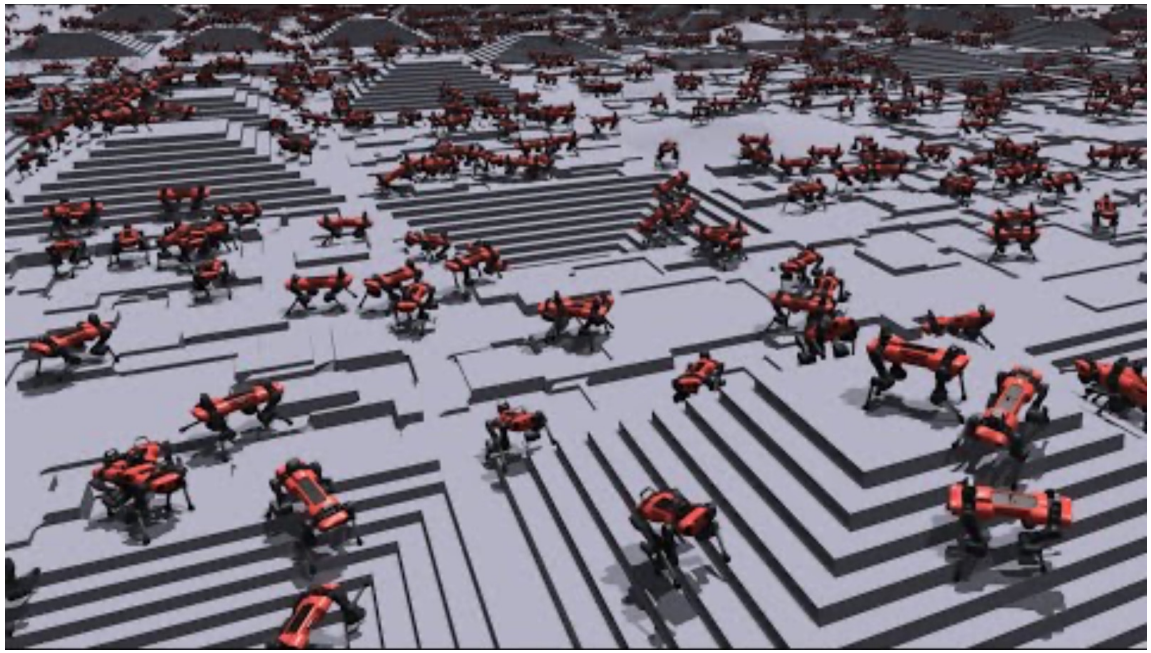
# Humanoid whole-body motion

Unitree H2

**Impressive!**
But really just stabilizing to one template human motion.

- Policy only does this one action
- Does not use vision (unaware of surroundings)
- Does not respond to language
- Not interactive or reactive to environment or people in any sense

# Quadruped Locomotion



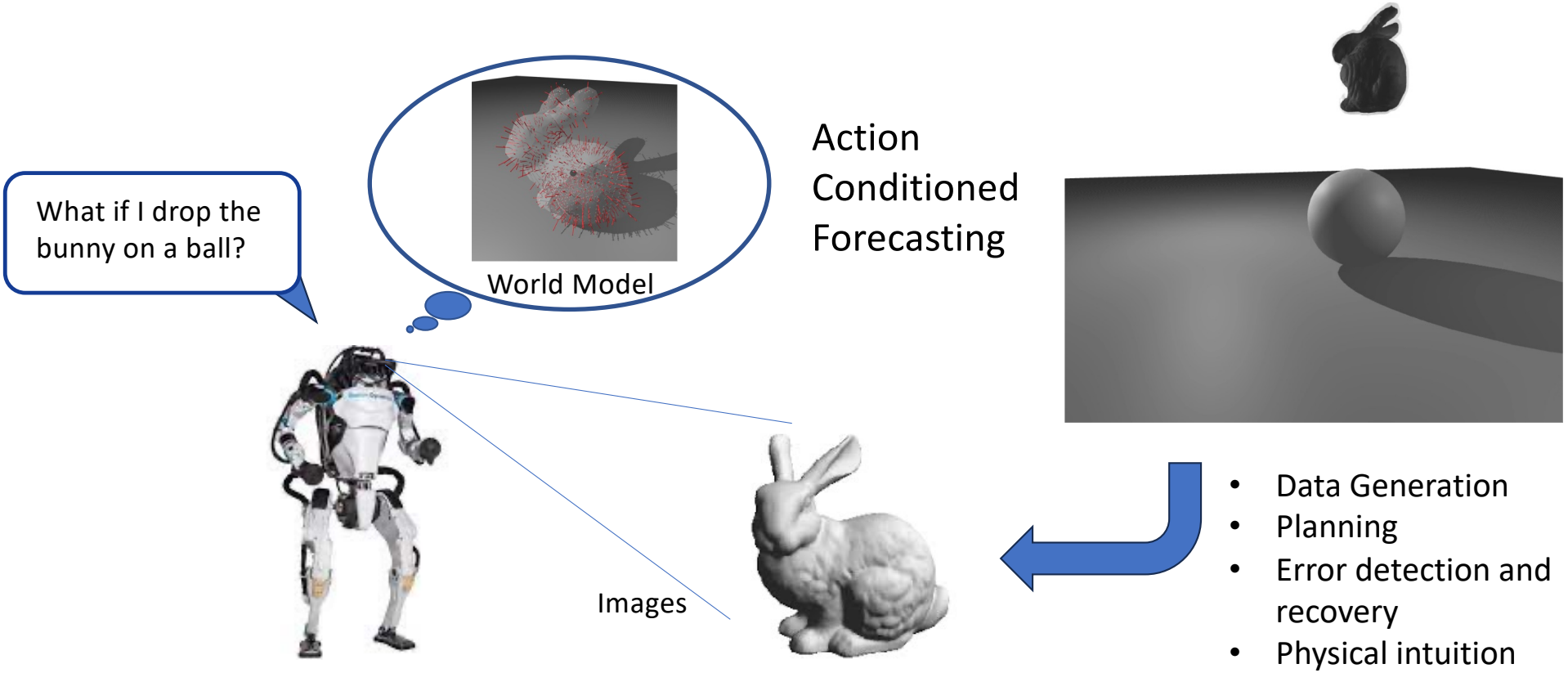
Prof. Marco Hutter, ETH Zurich

“Pure RL” no human data at all

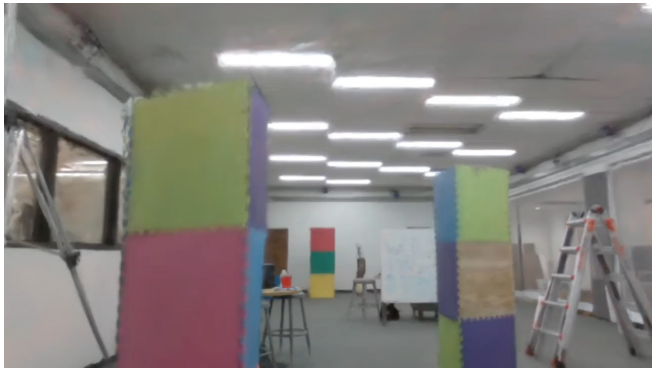
# Roadmap

- Examples of AI-Driven Robot Autonomy
- Robot autonomy policies
  - The robot autonomy stack
  - Imitation Learning (IL)
  - Reinforcement Learning (RL)
- World Models, 3D Models, and Simulators
- Challenges and opportunities

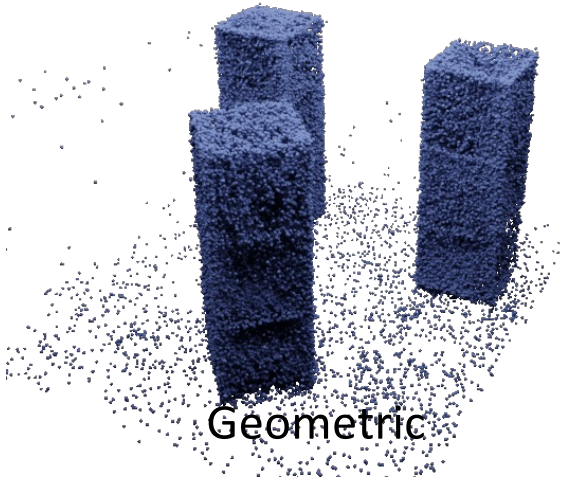
# World Models for Forecasting, Decision Making, Data Collection, Performance Analysis



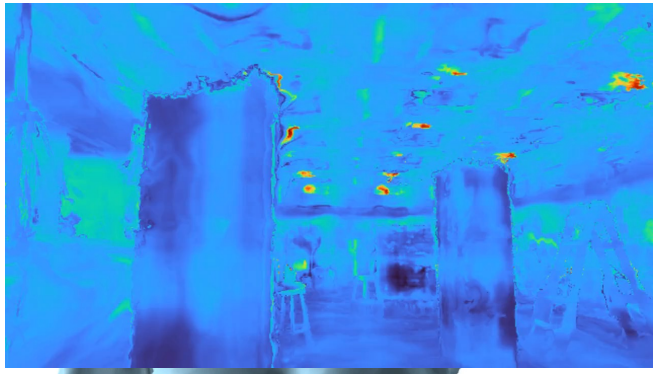
# 3D World Models that are…



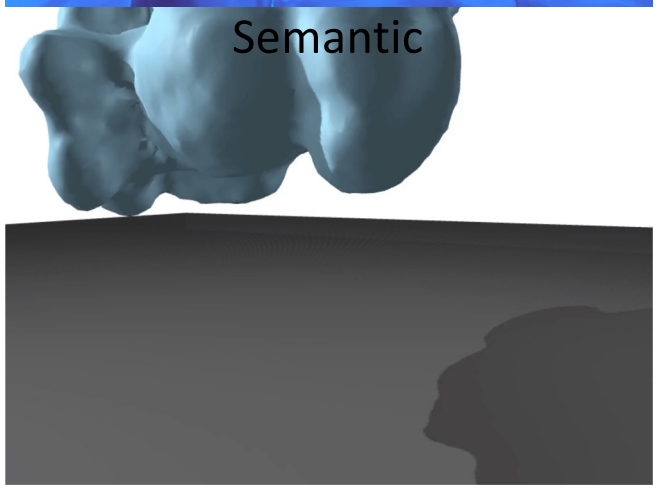
Visual



Geometric



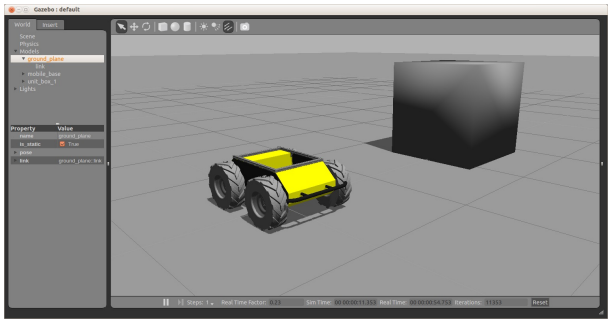
Semantic

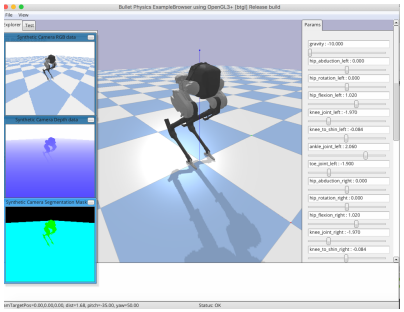


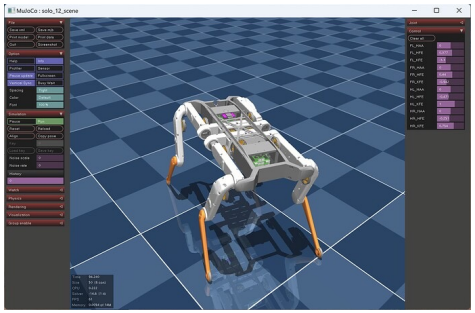
Dynamic

# Robot Simulators

Hand engineered based on classical physics and rigid body geometry.
No real world data or machine learning.







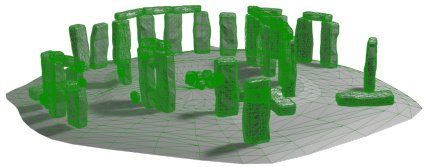
GAZEBO

BULLET PHYSICS LIBRARY

MuJoCo
Advanced physics simulation

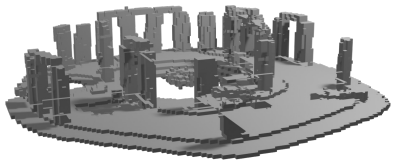
**Many, many more:** IsaacLab, Drake, Webots, DoJo, TiaChi …

# 3D geometry representations in robotics

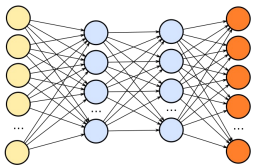


Mesh

Point cloud



Voxel grid

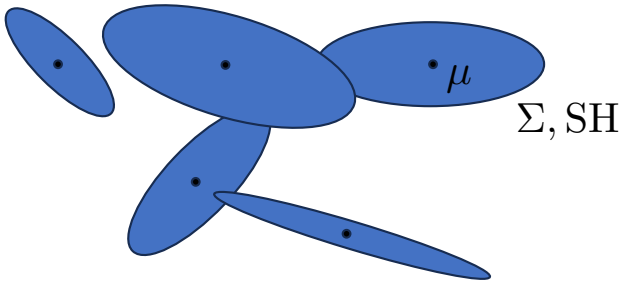


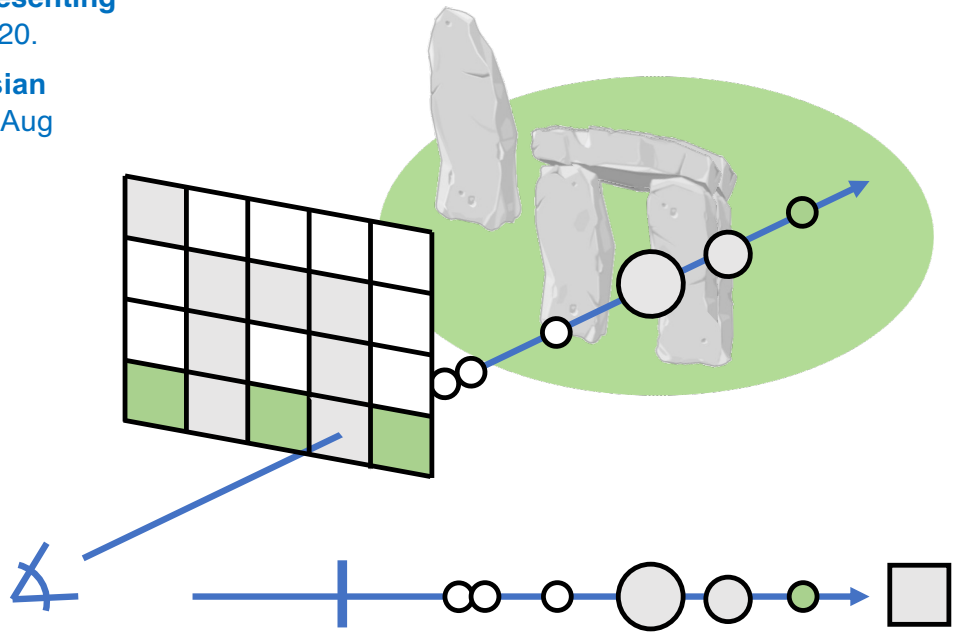
Neural representations

# 3D Gaussian Splatting (GSplat)

Mildenhall*, Srinivasan*, Tancik*, Barron, Ramamoorthi, Ng. **Representing Scenes as Neural Radiance Fields for View Synthesis**. ECCV '20.

Kerbl, B., G. Kopanas, T. Leimkühler, and G. Drettakis. **3D Gaussian Splatting for Real Tim Radiance Field Rendering**. SIGGRAPH, Aug 2023.



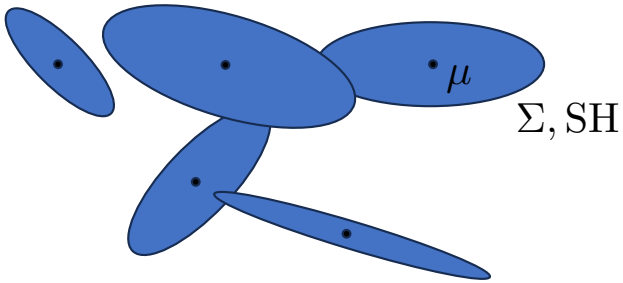



$$(\mathbf{x}, \mathbf{v}) \mapsto \alpha, \mathbf{c}$$

3D point view dir opacity radiance

# 3D Gaussian Splatting (3DGS)

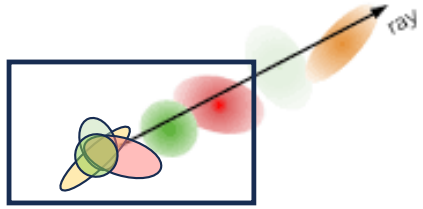
Kerbl, B., G. Kopanas, T. Leimkühler, and G. Drettakis. **3D Gaussian Splatting for Real Tim Radiance Field Rendering**. SIGGRAPH, Aug 2023.



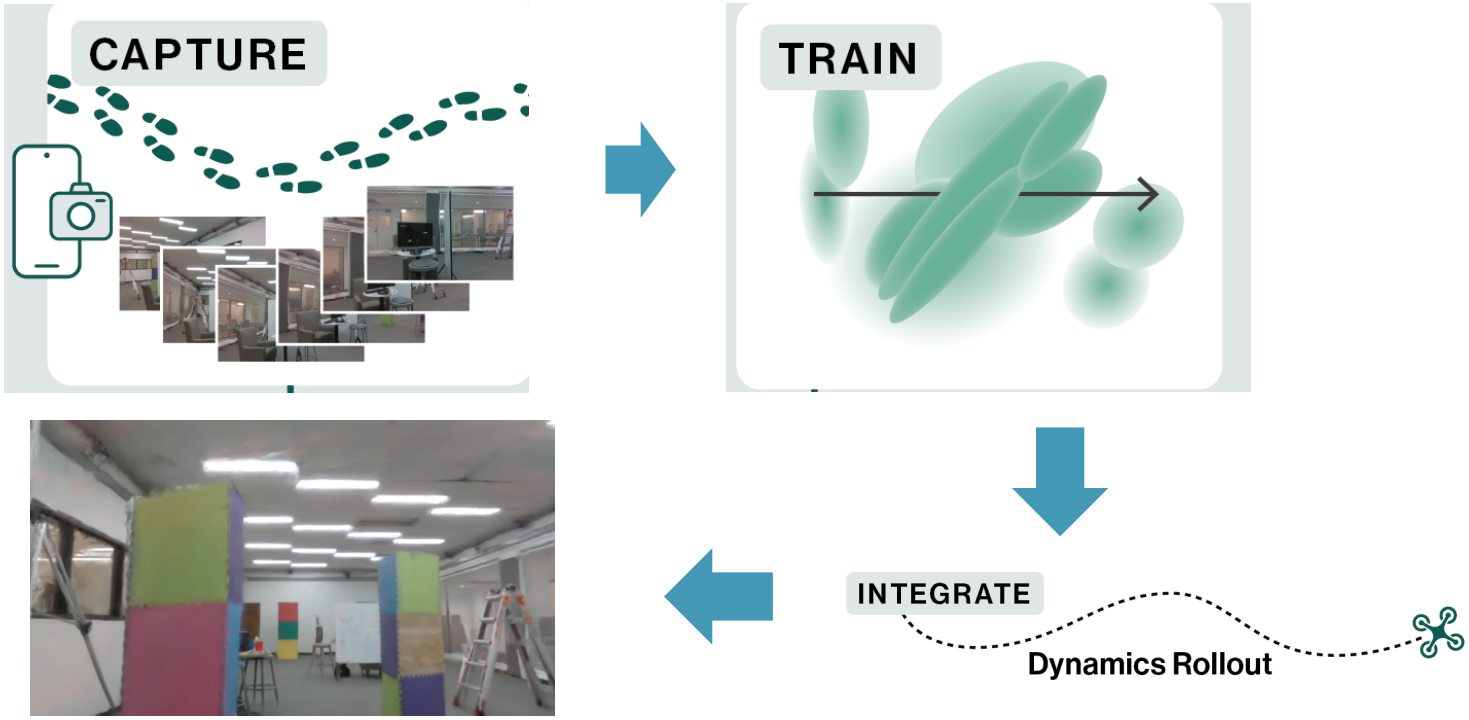
$$(\mathbf{x}, \mathbf{v}) \mapsto \alpha, \mathbf{c}$$

3D point  view dir  opacity  radiance

3DGS Rasterized Rendering:

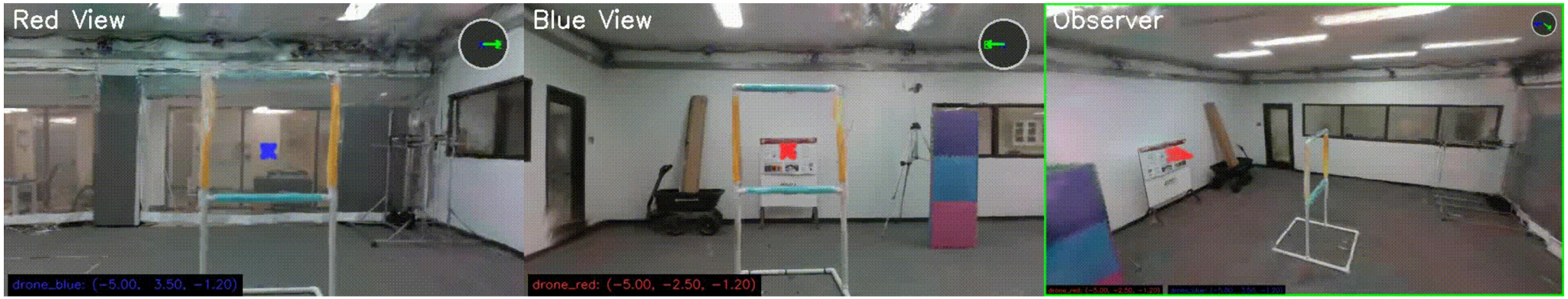


# Flying in Gaussian Splats (FiGS)




We build a 3D scene model (a Gaussian Splatting model) from images captured from a regular camera, integrate and drone dynamics model, and obtain a high-fidelity flight simulator we call FiGS.

# Multi-Drone FiGS

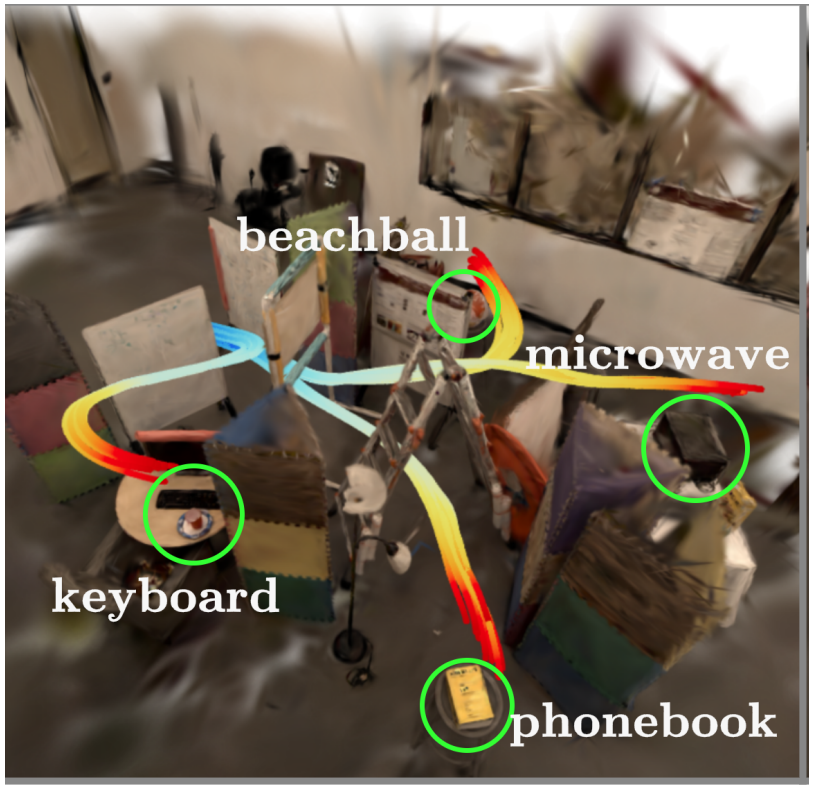



- Rendering from multiple agent viewpoints simultaneously
- Each agent views the other agent's motion in its simulated video feed

# Embedding Language: “Microwave”, “Penguin”

- We embed language codes (CLIP) so we can query the 3D scene model with a text prompt.

# Language Labeled Navigation Data




- FiGS with language embeddings lets us collect policy training data consisting of images, language labels, and robot actions.
- We can then train robot navigation policies that can take a human text command (“keyboard”) and navigate the robot to that object in the real world.

# Walking in Gaussian Splats (WiGS)

- We integrate with a humanoid dynamics simulator for a full visual-dynamic-semantic humanoid world model.

# Roadmap

- Examples of AI-Driven Robot Autonomy
- Robot autonomy policies
  - The robot autonomy stack
  - Imitation Learning (IL)
  - Reinforcement Learning (RL)
- World Models, 3D Models, and Simulators
- Challenges and opportunities

# Challenges and Opportunities

Current IL paradigm: We need 100Ms of high-quality demonstrations

(we're not even close)

- Better human operators? More operators?
- Better tele-op interfaces?

**Learn from YouTube and other large-scale data sources**

**Better world models**

**Broad-context self-evaluation models**

**Policies that learn from their own experience**